

Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 4

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

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Work Done in week 4:

1. Reusing the Week 2 Patch Dataset

No new preprocessing this week. The same patch set developed in Week 2 was loaded straight: 3,222 defect patches and 4,371 defect-free patches. For fabric type, each filename was again used to extract the fabric type to build up per-fabric texture references.

2. Shared Evaluation Framework

A common set of parameters and helper functions were first agreed on before implementing the new descriptors, in such a way that each descriptor will be assessed under the same conditions:

The common scale for all descriptors was set to be a fixed window size of 32×32 pixels, each having a stride of 16.

Windows with near zero or blank windows (standard deviation is less than a threshold of 5.0) were not included in the creation of reference textures because they are mostly background or border areas and do not represent actual fabric texture.

A shared conversion function was defined for the deviation to mask, to normalize the deviation map and Otsu's thresholding was used to obtain a binary predicted mask.

To ensure that all descriptors' outputs are evaluated in the same manner, a shared confidence score and metric calculation function (IoU, precision, and recall, F1) were defined.

3. LBP Baseline (Carried Over from Week 3)

This notebook re-introduces the multi-radius uniform LBP descriptor from week 3 for a more comprehensive comparison of all descriptors. A per-fabric-type reference histogram was constructed from patches that are free of defects, and approximately 27,300 blank windows were not used for constructing the reference.

4. GLCM (Gray-Level Co-occurrence Matrix) Descriptor

Implemented a new GLCM-based descriptor:

The co-occurrence matrix was calculated for each window, which was quantized to 8 gray levels.

Four Haralick features (contrast, energy, homogeneity, correlation) were calculated at distance 1 for four angles (0° , 45° , 90° , 135°) and then averaged.

A reference (mean and standard deviation of the feature vector) was created for each fabric type from defect-free patches and a standardized (z-score based) distance was calculated based on the deviation from this reference.

5. Gabor Filter Bank Descriptor

There were 12 filters in the Gabor filter bank, which had 4 orientations and 3 frequencies. The filter was applied to all the banks and each patch was filtered once and the response of the filter in each window was averaged to obtain the feature vector. Just like the GLCM, a reference of the fabric types was also built using defect-free samples of fabric, and the distance from the reference was also calculated using a standardized method.

6. Combined GLCM + Gabor Descriptor

To see if texture and frequency information can be combined to improve detection, a fourth descriptor was generated by simply forming a concatenation of the GLCM feature vector and the Gabor feature vector for each window, and making a new reference statistics and deviation map for the combined descriptor.

7. Validation Sample and Comparison

To be able to test all the descriptors on the same images, a fixed and seeded sample of 10 defects containing patches was chosen. This sample was tested with all four descriptors and the masks predicted by these descriptors were evaluated with respect to the ground-truth masks, using IoU, precision, recall, and F1. A visual comparison was also generated, which shows the overlay of the forecasted defect

when compared to the ground truth for the sample patch shown for each descriptor.

8. Descriptor Comparison and Baseline Selection

Average metrics of the validation sample were calculated per descriptor. The average F1 score was higher for LBP (0.136) than for the other descriptors, GLCM and the combined descriptor had higher recall (0.068, 0.086 respectively). This was used to select the single-scale baseline descriptor LBP to proceed to the next stage of development.

Challenges Faced

1. Selecting the correct texture features.

The most prominent difficulty encountered was the selection of features which would effectively differentiate defective fabric from normal fabric. Different methods are capturing different texture patterns, so we explored the use of LBP, GLCM, Gabor, a combination of GLCM and Gabor to understand the contribution of each method to the defect detection.

2. Selecting Suitable Parameters.

Another difficulty was setting the parameters for GLCM and Gabor filters. The features extracted from an image can vary with distance, angle, orientation and frequency. The choice of suitable values was crucial for the ability to capture texture information required to identify the defects.

3. Handling various kinds of fabrics.

Fabrics vary in the texture and normal variations in one fabric may be different to another. This posed the problem of developing a standard to identify defects. For this, we took texture references for each fabric as much as possible.

4. Defects detection without misclassification of defect-free regions.

A few common fabric patterns may look like flaws and aren't always reliably distinguished. The deviation maps and thresholding were used to find the areas that are different from the expected texture. Selecting an appropriate threshold was still significant to realize meaningful defect predictions, though.

5. Combining GLCM and Gabor Features.

The problem was merging GLCM and Gabor, they have different definitions of texture. In GLCM, the emphasis is on relationships of neighbouring pixel intensities and Gabor filters are used to extract patterns in different directions. By using them together, both types of information could be used for defect detection.

6. Fairly comparing different methods.

The second difficulty was to make sure that the various feature extraction techniques were applied to comparable test problems. We used the same validation sample and evaluation metrics to compare their performance. This allowed us to see the difference between the methods more clearly.

7. Measurement of the degree of accuracy for defect detection.

The detection of a defect is not just a matter of finding that there is a region in the image that is unusual; it is also important that this region is found at the correct location. We used metrics such as IoU, Precision, Recall, and F1-score to compare the predicted defect regions with the actual defect masks and assess the performance of each method.

Work to be done next week

1. Multi-Scale Feature Extraction: Apply the current texture descriptors (LBP, GLCM, Gabor) at various window sizes, various radii and various frequencies. It will be more useful for capturing both fine and coarse fabric texture patterns that may be overlooked using a single scale.
2. Fusion of Multi-Scale Features: Create maps of deviations between different scales and fuse them, for example, through weighted summation

or maximum response. This will aid in the detection of texture anomalies of different size and appearance in various fabric samples.

3. Adaptive Parameter Selection: Design and develop an adaptive parameter selection for varying window size and other parameters based on the texture characteristics of the seven fabric types. The texture periodicity can be estimated by applying autocorrelation or peak detection on the Fast Fourier Transform (FFT) to choose the proper parameters for each type of fabric.
4. Make 2 pipelines, Fixed and Adaptive. The first will be applied to every fabric type with predetermined parameters, whereas the second will adjust the parameters of the fabric based on the estimated texture characteristics of each fabric.
5. Evaluate both pipelines on the validation data set and compare the performance in terms of anomaly detection. Compare adaptive parameter selection to see if it is improving the detection accuracy and also gives better detection results for different texture patterns of fabrics.